



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Digital Learning Handout

Part A: Content Design

Course Title	Distributed Machine Learning
Course No(s)	AIMLC ZG515
Credit Units	4
Credit Model	3-1-0
Instructors	Prof Chandrasekhar V, Prof Mohammed Nazaar M, Prof Eranki L N Kiran Kumar
Version No:	1.0
Date:	28/02/2025

Course Description:

Introduction to parallel and distributed models of computation: Scalable frameworks to parallelize machine learning algorithms, Data and computation heterogeneity, Data parallelism vs Model parallelism, Challenges: consistency, fault tolerance, communication, resource management, programming models; Distributed ML algorithms: K-means, DBSCAN, Distributed association rule mining: FDM, Linear and logistic regression; Distributed DL Algorithms: Gradient descent techniques for empirical risk minimization, SGD in Neural Network Training and its convergence analysis, Distributed Synchronous SGD, Asynchronous SGD, Hogwild, Local-update SGD, Decentralized SGD, Overlap SGD, Quantized SGD, Adacomm Vs Distributed SGD, Elastic Averaging, AdaSyn, AdaQuant; Federated learning: Privacy and security in federated learning; Hyper parameter optimization; In depth case studies of a few algorithms.

Course Objectives

No	Course Objective
CO1	To expose learners to the integration of machine learning algorithms with modern computing systems
CO2	To explain the computational performance and scalability of these algorithms on platforms like multi-core CPUs, GPGPUs, clusters, and constrained devices.
CO3	To highlight the impact of performance improvement techniques on domain-specific (ML) quality attributes.
CO4	To provide a systems-level understanding of ML/DL, drawing from foundational ML algorithm knowledge and focusing on computing systems that support ML applications.

Text Book(s):

T1	Distributed Machine Learning with Python Accelerating model training and serving with distributed systems by Guanhua Wang, packet publishers.
T2	Distributed Machine Learning Patterns by YUAN TANG, Manning publishers.
T3	Federated Learning with Python, Kiyoshi Nakayama, PhD, Packet publishers.

Learning Outcomes: Students will be able to

LO1	Learners will understand and articulate how parallel/distributed ML algorithms leverage standard platforms for ML to enhance performance.
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LO2	Learners will gain hands-on experience implementing parallel/distributed ML algorithms on clusters and constrained/Small-Form-Factor devices.
LO3	Learners will be able to argue and demonstrate the systems-level performance across a broad spectrum of parallel/distributed ML algorithms.

Modular Content Structure

1. Foundations of Distributed ML

- Overview of Distributed Systems
- Introduction to Machine Learning in Distributed Systems
- Shift from single-machine to distributed ML models.
- Role of multi-core CPUs, GPUs, clusters, and cloud platforms
- How hardware advancements enable scalable ML solutions.

2. Scalability in Distributed ML

- Overview of Scalable Frameworks
- Data and Computation Heterogeneity

3. Distributed Algorithms for ML

- Data Parallelism
- Model Parallelism
- Distributed ML Challenges
- Programming Models for Distributed ML
- Core Distributed ML Algorithms: Distributed Clustering, Distributed Association Rule Mining
- Distributed Linear and Logistic Regression
- Advanced Distributed Deep Learning Techniques (Gradient Descent)
- Advanced SGD Techniques

4. Optimizing ML in Constrained Environments

- Introduction to Federated Learning
- Security Considerations
- Hyperparameter Optimization in Federated Settings
- Comparative Analysis: Adacomm vs. Distributed SGD

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Introduction to Distributed Machine Learning	• Overview of Distributed Systems • Definitions and key differences: Parallel vs. Distributed Computing. • Importance of distributed systems in ML. • Introduction to Machine Learning in Distributed Systems: • Scalability challenges in ML. • Overview of distributed ML applications and benefits.	T1 and lecture notes
2	Scalable Frameworks for ML	• Scalable Frameworks Overview • Introduction to Hadoop, Spark, and other big data technologies.	T1 and lecture notes





		• Comparison and use cases in ML. • Data and Computation Heterogeneity • Handling heterogeneous data sources. • Adapting ML algorithms for heterogeneous computing resources.	
3	Data Parallelism	• Basics of Data Parallelism: • Concept and architectural overview. • Splitting data across multiple processors. • Implementing Data Parallelism: • Practical examples: Batch processing in deep learning. • Challenges and solutions in data parallelism.	T1 and lecture notes
4	Model Parallelism	• Basics of Model Parallelism: • Concept and architectural overview. • Splitting model parameters or computations. • Implementing Model Parallelism: • Practical examples: Large-scale neural networks. • Challenges and solutions in model parallelism.	T1 and lecture notes
5	Distributed ML Challenges - Part 1	• Consistency Models in Distributed Systems: • Definitions and importance. • Eventual vs. strong consistency. • Fault Tolerance Mechanisms in Distributed ML: • Strategies for handling failures. • Checkpointing and recovery methods.	T1 and lecture notes
6	Distributed ML Challenges - Part 2	• Communication Overhead in Distributed ML: • Identifying and minimizing communication bottlenecks. • Efficient data serialization and transmission techniques. • Resource Management and Allocation: • Balancing load and computational resources. • Dynamic resource allocation strategies.	T1 and lecture notes
7	Programming Models for Distributed ML	• Distributed Programming Models Overview: • MapReduce, Spark RDDs, and beyond. • Suitability for different types of ML algorithms. • Case Studies: • Real-world applications and their architectural choices. • Lessons learned from deploying distributed ML systems.	T1 and lecture notes
8	Review	Review of contact session 1 to 7	
9	Core Distributed ML Algorithms	• Distributed Clustering Algorithms: • K-means and DBSCAN adaptations for distributed environments.	T1 and lecture notes





		• Synchronization and convergence challenges. • Distributed Association Rule Mining (FDM): • Implementing Frequent Directional mining in distributed settings. • Efficiency and scalability considerations.	
10	Distributed Linear and Logistic Regression	• Adapting Regression Models for Distributed Systems: • Partitioning strategies for datasets. • Parallelizing computations for gradient descent. • Challenges and Solutions: • Dealing with data skewness and imbalance. • Ensuring model convergence across distributed systems..	T1 and lecture notes
11	Advanced Distributed Deep Learning Techniques	• Gradient Descent in Distributed Environments: • Empirical risk minimization strategies. • Variants of gradient descent: Batch, Mini-batch, and Stochastic. • SGD in Neural Network Training: • Challenges in distributed SGD implementations. • Techniques for improving convergence and accuracy..	T1 and lecture notes
12	Advanced SGD Techniques	• Exploring Advanced SGD Variants: • Synchronous vs. Asynchronous SGD: Benefits and drawbacks. • Specialized techniques: Hogwild, Local-update SGD, Decentralized SGD. • Quantization and Compression in SGD: • Quantized SGD for reduced communication overhead. • Compression techniques for efficient data exchange.	T1 and lecture notes
13	Federated Learning - Foundations	• Introduction to Federated Learning: • Core concepts and architectures. • Privacy-preserving machine learning. • Security Considerations: • Threat models in federated learning. • Secure aggregation and multi-party computation.	T3 and lecture notes
14	Federated Learning - Advanced Topics	• Hyperparameter Optimization in Federated Settings: • Challenges of hyperparameter tuning in distributed environments. • Strategies and tools for effective optimization. • Case Studies: • Success stories and lessons learned from	T3 and lecture notes





		federated learning projects. • Analysis of deployment strategies and outcomes.	
15	Special Topics in Distributed ML	• Comparative Analysis: Adacomm vs. Distributed SGD: • Theoretical underpinnings and practical implications. • Performance, scalability, and applicability.	T3 and lecture notes
16	Review	Review of contact session 9 to 15	

Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc...) number of lab exercises needed, etc.

1. Lab work: 14
2. Project work: 1
3. Case Study: 0 Webinars
4. Simulation: 0
5. Work Integrated Learning Assignment- 1 Assignment
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of Distributed ML using state of art tools

Scope of Experiential Learning Component:

Programming language - Python

Tools and libraries: Docker, pyspark, tensorflow, torch, pickle, os, mrjob, horovod, numpy, matplotlib, tensorflow-federated

Lab Infrastructure:

Online/ Open source/Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Setting up a distributed computing environment using Docker and exploring basic distributed computing operations with Python's multiprocessing module	1
2	Hands-on exercise with Apache Spark to parallelize data processing tasks, using PySpark to perform simple data aggregations on a distributed dataset.	2
3	Implementing data parallelism for a basic ML model training task using TensorFlow or PyTorch distributed.	3
4	Coding exercise to implement model parallelism for a large neural network using PyTorch's nn.parallel API	4
5	Simulating fault tolerance in a distributed setting using a simple Python script to demonstrate checkpointing and recovery.	5
6	Exercise on optimizing communication in a distributed TensorFlow	6





	training job, using TensorFlow's tf.distribute.Strategy.	
7	Implementing a MapReduce job using Hadoop or Spark for a large-scale data processing task relevant to ML	7
8	Coding distributed K-means clustering using PySpark's MLlib	9
9	Implementing distributed logistic regression using Spark MLlib	10
10	Hands-on with distributed SGD training of a deep learning model using Horovod or TensorFlow's distributed strategies	11
11	Experimentation with different SGD variants using a simple Python framework to observe convergence rates and performance.	12
12	Building a simple federated learning system using TensorFlow Federated (TFF) to train a model across decentralized datasets	13
13	Advanced federated learning project focusing on optimizing model performance across distributed devices.	14
14	Implementing and comparing the performance of Adacomm and Distributed SGD on a distributed dataset to understand their practical applications and limitations.	15

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Assignment	Online	15%	3 to 4 weeks	To be announced
	Project	Online	30 %	6 weeks	To be announced
EC - 2	Mid-Semester Test	Closed Book	25%	2 hours	20-06-2026 (EN)
EC - 3	Comprehensive Exam	Open Book	30%	2 ½ Hours	05-09-2026 (EN)

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.





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[EC3 Guidelines](#)

4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

